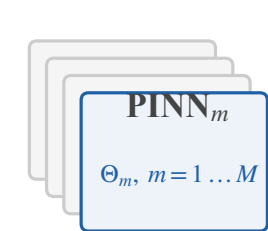


1 Deep ensemble of differentiable surrogates



independently retrained
(M random seeds)

design Jacobian
 $\partial \mathbf{r} / \partial \mathbf{g}_k$ (free autodiff)

per-component spread
across seeds

2 Solver-free sign-agreement gate

sign-agreement score $a_k \in [0, 1]$

impedance gradient

seed-stable



TRUSTED: free autodiff

radiation gradient

seed-unstable



FLAGGED: 1 full-wave solve

$\tau = 0.9$

3 Cost-optimal hybrid gradient

hybrid
gradient

$\nabla_g \mathcal{L}$

- free autodiff on trusted
- finite-difference on flagged only

AUC **0.91** predicting the full-wave verdict
on an external transfer-matrix benchmark

2 vs 11 FD checks to catch the
sign-wrong gradients (gate vs $|\nabla|$)

A solver-free deep-ensemble gradient-reliability gate: trust free autodiff where seeds agree, verify only where they do not.